


# CMIS Event Management System RFP

## Group 5

**Sri Akhil Arigela - 836001287**

**Gauri Babele - 937006611**

**Sonali Dash - 837003064**

**Yashas Suresh - 737002366**

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# FOREWORD

At Mays Business School, Texas A&M University, the Management Information Systems Program is dedicated to developing leaders who use technology and data to drive innovation and make a difference. The Council for the Management of Information Systems (CMIS) plays a central role in this mission by connecting students, faculty and corporate partners through a wide range of events and experiential learning opportunities.

As CMIS continues to expand its programs and corporate partnerships, the need for a unified and data-driven platform to manage its growing event operations has become clear. The CMIS Event Management System (EMS) will serve as a key step toward that goal, providing a single, efficient and central intelligent system to support planning, coordination and reporting across all CMIS activities.

This initiative reflects Mays Business School's commitment to continuous improvement, operational excellence and digital transformation. By investing in a modern integrated event management solution, CMIS is creating the foundation for a more connected, innovative and impactful experience for students, sponsors and alumni that embodies the values and spirit of Texas A&M University.

# INTRODUCTION

At Mays Business School at Texas A&M University, the Management Information Systems (MIS) Program is dedicated to developing leaders who combine business acumen with technological expertise to drive innovation and impact. The Council for the Management of Information Systems (CMIS) plays a pivotal role in this mission by serving as the bridge between academia and industry by connecting students, faculty, alumni and corporate partners through a wide range of events, career fairs and professional development programs.

As CMIS expands its outreach and partnerships, the growing number and complexity of events have highlighted the limitations of existing processes. Currently, event registration, scheduling, sponsor coordination and feedback collection are handled through a fragmented mix of emails, spreadsheets and online forms. This disjointed approach consumes significant faculty and student time, causes data inconsistencies and restricts the ability to analyze engagement or measure the long-term impact of CMIS activities. These challenges highlight the need for a unified, intelligent and data-driven system that can manage operations efficiently while enhancing stakeholder experience.

The proposed CMIS Event Management System (EMS) envisions a comprehensive cloud-based platform that integrates every stage of the event lifecycle from planning and registration to execution and evaluation into one cohesive digital environment. The EMS will automate key workflows such as confirmations, attendance tracking and reporting to reduce administrative overhead and ensure data accuracy across all events. By transforming raw participation data into actionable insights, the system will empower CMIS leadership to assess engagement trends, sponsorship outcomes and overall return on investment.

Beyond efficiency, the EMS reflects CMIS's broader vision of fostering collaboration, accessibility and innovation. It will offer role-based dashboards that provide tailored experiences for students, sponsors, faculty and administrators while maintaining compliance with Texas A&M University's data security and accessibility policies including FERPA and WCAG 2.1 standards. The platform will serve as the digital backbone of CMIS operations by creating a living repository of event intelligence that supports predictive analytics, alumni engagement and future integrations such as mentorship and research collaboration modules.

Through this Request for Proposal (RFP), the student body of Mays Business School seeks approval to design, develop and implement the CMIS Event Management System in partnership with a qualified development team. This initiative represents more than just a technological enhancement; it is a strategic investment in CMIS's mission to build stronger connections between students and industry, streamline operations and deliver a best-in-class digital experience that upholds the Aggie values of excellence, leadership and innovation.

# 1. SCOPE

The CMIS Event Management System (EMS) project aims to design and develop a dedicated website that will serve as the central hub for all event-related activities within the Council for the Management of Information Systems (CMIS) at Mays Business School, Texas A&M University. The goal is to create a seamless and accessible platform that manages the complete event lifecycle from announcements and registration to reporting and feedback while improving coordination, transparency and engagement among students, sponsors, faculty and administrators.

Presently, CMIS relies on fragmented tools and manual processes that cause inefficiencies, communication gaps and limited visibility into event performance. These challenges restrict the ability to make timely, data-driven decisions and to measure the success of initiatives. Through this project, CMIS aims to build a unified and intelligent system that automates key operations, strengthens collaboration and enhances the overall stakeholder experience.

The EMS will include a website that highlights upcoming events, announcements and sponsor information along with a secure portal where students, faculty and sponsors can manage profiles, registrations and participation. It will include built-in tools for scheduling, attendance tracking, communication and document sharing, as well as administrative dashboards that help CMIS staff monitor event metrics and generate reports. An important feature will be the automatic generation of unique QR codes for all registered users to enable secure and efficient check-ins managed by an approved third-party service provider.

To further enhance operational efficiency, the EMS will feature real-time analytics and reporting capabilities that allow administrators and faculty to track participation rates, evaluate sponsor engagement and identify areas for improvement. The inclusion of visual dashboards and automated summary reports will enable leadership to measure event impact and optimize planning for future initiatives. The system's data-driven design will ensure that every event contributes to institutional insights, aligning with CMIS's broader objective of promoting continuous learning and performance improvement.

The system will be designed with a user-centred approach, ensuring intuitive navigation, mobile responsiveness and compliance with accessibility and security standards. The interface will provide clear pathways for registration, approval and feedback, reducing the learning curve for new users and minimizing dependence on manual support. By aligning with Texas A&M University's IT governance, branding and data privacy policies, the EMS will ensure long-term sustainability and scalability, creating a foundation for future modules and digital transformation initiatives across CMIS.

The project will focus exclusively on the design and development of the Event Management System (EMS) as the central hub of CMIS's operations. While CMIS's broader ecosystem includes related modules such as the Student Career Portal, Case Competition Platform, Industry Speaker Exchange, Mentorship Marketplace, Research Collaboration Hub, Alumni Engagement Portal and Analytics Dashboard, this project will refrain from developing or maintaining those components. The EMS will be designed to stay compatible for future integration with these systems, but their implementation is outside the current scope. Additionally, the project will not involve the creation of unrelated websites, standalone mobile applications or integration.

Overall, this initiative aims to deliver a secure, user-friendly and data-driven solution that simplifies event management, enhances stakeholder engagement and strengthens CMIS's mission of connecting students and industry partners through technology and collaboration.

# 2. Normative References

The following standards and documents are referenced within this Request for Proposal (RFP). Portions of these documents are incorporated by reference and their provisions constitute essential requirements of this document.

For dated references, only the specific edition cited shall apply. For undated references, the latest edition, including any amendments or revisions, shall apply.

1. **ISO/IEC/IEEE 15288:2015** — Systems and Software Engineering — System Life Cycle Processes

   This standard provides a framework for system life cycle processes, defining a common set of terminology and activities to guide the development, implementation and maintenance of complex systems such as the Event Management System (EMS). It ensures that the system follows recognized best practices from concept to retirement.

2. **ISO/IEC/IEEE 12207:2017** — Systems and Software Engineering — Software Life Cycle Processes

   This standard establishes processes for managing the software component of the system, including acquisition, development, operation, maintenance and disposal. It ensures software quality, traceability and compliance throughout the EMS development and deployment phases.

Together, these standards provide the foundational framework for defining requirements, processes and quality assurance practices throughout the life cycle of the University's Event Management System.

# 3. Terms, Definitions, and Abbreviated Terms

## 3.1 Terms and Definitions

For the purposes of this document, the terms and definitions given in ISO/IEC/IEEE 15288, ISO/IEC/IEEE 12207 and the following apply specifically to the CMIS Event Management System (EMS).

The EMS is defined as the University's centralized digital platform designed to support the end-to-end planning, coordination, execution and analytics of institutional events across academic, administrative and industry engagement domains.

Normative references for terminology are also available at:

- ISO Online Browsing Platform: `http://www.iso.org/obp`
- IEC Electropedia: `http://www.electropedia.org`
- IEEE Standards Dictionary Online: `http://dictionary.ieee.org`

### 3.1.1 Acquirer

The stakeholder (e.g., University administration or procurement office) responsible for acquiring or approving the Event Management System for institutional use.

*Example:* Texas A&M CMIS acts as the acquirer of the EMS solution.

### 3.1.2 Attribute

A measurable or descriptive property of an EMS entity, such as event date, venue capacity, or registration status. Attributes define how data is captured and used within the system.

### 3.1.3 Baseline

A formally approved version of the EMS configuration (e.g., system setup, workflow, or data model) that serves as the reference point for future updates or audits.

### 3.1.4 Business Requirements Specification (BRS)

A structured collection of high-level business needs that define what the EMS must achieve, including stakeholder engagement, scheduling automation and reporting requirements.

### 3.1.5 Concept of Operations (ConOps)

A verbal and visual description of how the EMS will be employed to plan and manage campus events, covering processes such as event creation, approval, registration, sponsorship, and feedback.

### 3.1.6 Condition

A qualitative or quantitative attribute that specifies when or how a requirement applies within EMS (e.g., "Registrations must close 24 hours before event start time").

### 3.1.7 Constraint

A limitation affecting EMS design or operation, such as university IT policies, data privacy standards (FERPA compliance), or event scheduling conflicts.

### 3.1.8 Context of Use

The environment in which the EMS operates — including users (students, faculty, administrators, industry partners), devices (desktop, mobile) and technical infrastructure (university servers, cloud hosting).

### 3.1.9 Customer

Any individual or organizational unit that uses or benefits from the EMS, including event organizers, student groups and external sponsors.

### 3.1.10 Derived Requirement

A secondary requirement inferred from primary EMS needs — e.g., from the need to "manage hybrid events," a derived requirement is "integration with Zoom or MS Teams APIs."

### 3.1.11 Developer

The internal or external technical team responsible for EMS design, development, testing and implementation.

### 3.1.12 Document

Any information artifact generated during the EMS life cycle — e.g., RFPs, system specifications, design blueprints, or user manuals.

### 3.1.13 Event

A scheduled university function (e.g., seminar, workshop, career fair, or competition) managed within the EMS, comprising metadata such as type, venue, time and participants.

### 3.1.14 Human Systems Integration (HSI)

The process of designing EMS interfaces and workflows that prioritize usability, accessibility and efficient interaction for diverse users.

### 3.1.15 Operator

A university staff member or system administrator responsible for managing and maintaining the EMS operations.

### 3.1.16 Requirement

A specific, verifiable statement defining what the EMS shall do — e.g., "The system shall prevent double-booking of venues."

### 3.1.17 Requirements Elicitation

The systematic process of gathering EMS needs from stakeholders such as event organizers, IT administrators and students through surveys, interviews and prototypes.

### 3.1.18 Requirements Traceability Matrix (RTM)

A structured mapping that links EMS requirements to their sources, design modules and testing criteria to ensure completeness and accountability.

### 3.1.19 Stakeholder

Any person or group that has an interest in or influence over the EMS — including students, faculty, administrators, IT services and industry partners.

### 3.1.20 System-of-Interest

The CMIS Event Management System (EMS) — encompassing its software, data repositories, interfaces, and operational processes within the university ecosystem.

### 3.1.21 User

An individual interacting directly with the EMS, including:

- **Student users** — registering or providing feedback
- **Faculty users** — hosting events or reviewing analytics
- **Administrators** — approving, scheduling, or reporting events

#### 3.1.22 Validation

The process of confirming that the EMS meets university expectations and stakeholder needs — "the right system has been built."

#### 3.1.23 Verification

The process of confirming that the EMS has been built correctly according to its requirements, design and technical specifications.

## 3.2 Abbreviated Terms

API - Application Programming Interface
BRS - Business Requirements Specification
ConOps - Concept of Operations
EMS - Event Management System
FERPA - Family Educational Rights and Privacy Act
HSI - Human Systems Integration
MIS - Management Information Systems
RTM - Requirements Traceability Matrix
SRS - Software Requirements Specification
StRS - Stakeholder Requirements Specification
SyRS - System Requirements Specification
TBD - To Be Determined
TPM - Technical Performance Measure
UI/UX - User Interface / User Experience

# 4. Conformance

## 4.1 Intended Usage

This section defines how the CMIS Event Management System (EMS) will demonstrate compliance with university standards, stakeholder expectations and ISO/IEC/IEEE 29148 principles.

The EMS will serve as a centralized data-driven platform that replaces fragmented workflows based on emails, spreadsheets and forms. It will ensure that all requirements are traceable, verifiable and measurable in alignment with CMIS's mission to connect students, faculty, alumni and industry partners. The system will enable planning, execution and evaluation of events in one unified environment while maintaining continuous validation of requirements through documentation, stakeholder reviews and audit trails.

## 4.2 Full Conformance

Full conformance with ISO/IEC/IEEE 29148 and 15288 standards will be maintained throughout the project lifecycle.

This will include the development of all required information items such as the Business Requirements Specification (BRS), Stakeholder Requirements Specification (StRS), System Requirements Specification (SyRS) and Software Requirements Specification (SRS) with proper version control and traceability.

The EMS will adhere to Texas A&M University IT policies on accessibility, data privacy and cybersecurity. Each requirement will be verified through inspection, testing or demonstration using a formal Requirements Traceability Matrix. Lifecycle governance will be monitored through the CMIS Change Control Board to ensure that all processes and deliverables meet institutional and technical quality standards.

## 4.3 Conformance to Processes

The EMS will comply with the three principal processes defined in the MSIS and CMIS RFP framework:

**Business or Mission Analysis Process:** Validates the strategic need for a unified digital platform to resolve data fragmentation and operational inefficiencies.

**Stakeholder Needs and Requirements Definition Process:** Converts stakeholder expectations into measurable, testable functional and non-functional requirements.

**System and Software Requirements Definition Process:** Formalizes detailed, verifiable system and software requirements aligned with university architecture, FERPA compliance and AWS deployment standards.

Conformance to these processes will be verified through iterative review cycles, traceability reports and pilot event validations conducted under faculty supervision.

## 4.4 Conformance to Information Item Content

All information items including BRS, StRS, SyRS and SRS will contain the required identification, front matter, definitions, references and acronyms as outlined in the MSIS document.

Each item will demonstrate traceability from the problem statement to the proposed solution and will align with CMIS's goals of automation, efficiency and stakeholder engagement.

Content reviews will confirm that each information item meets the standards of clarity, completeness and verifiability established by ISO/IEC/IEEE 29148.

## 4.5 Tailored Conformance

### 4.5.1 Processes

Tailoring will emphasize scalability, clarity and resource efficiency within the academic environment. Iterative development will be used to refine requirements based on pilot event feedback. All process deliverables will remain traceable and auditable while being simplified for use by students and faculty. Compliance and verification cycles will be aligned with the university academic calendar to ensure timely development and deployment.

### 4.5.2 Information Items

Each information item will be tailored to the specific scope of the CMIS Event Management System.

The Business Requirements Specification will define organizational value and performance goals. The Stakeholder Requirements Specification will capture user expectations for speed, accuracy and usability. The System Requirements Specification will describe system interfaces and integration with platforms such as AWS QuickSight and TAMU SSO. The Software Requirements Specification will define functional logic, validation mechanisms and data security parameters.

# 5. CONCEPTS

## 5.1 General

The CMIS Event Management System (EMS) is designed to serve as a unified digital platform that streamlines the planning, registration, coordination and reporting of events organized under the Council for the Management of Information Systems (CMIS) at Mays Business School. This section defines the underlying concepts and principles that guide the requirements engineering process for the EMS. The goal is to ensure that every requirement reflects stakeholder intent, institutional policy and measurable business outcomes.

### 5.1.1 Requirements Fundamentals

**General** The EMS requirements are structured according to international standards (ISO/IEC/IEEE 29148 and 15288) to maintain traceability, clarity and verifiability throughout the project life-cycle. Each requirement must directly address CMIS's mission to enhance engagement, automation, transparency and data integrity across events.

**Stakeholders** Stakeholders include students, faculty, sponsors, administrators, alumni and Texas A&M University IT Services.

- Students require intuitive registration, timely confirmations and easy access to event schedules.
- Faculty and Administrators need tools for approvals, tracking attendance, generating reports and ensuring compliance.
- Sponsors and Industry Partners seek visibility into participation, branding opportunities and engagement outcomes.
- IT Services ensures adherence to university cybersecurity, FERPA compliance and WCAG 2.1 accessibility.

**Transformation of Needs into Requirements** Stakeholder needs are translated into actionable functional and non-functional requirements through interviews, process mapping and validation workshops. For example, the need for accurate attendance is expressed as a functional requirement for automated QR code check-ins. Similarly, the desire for inclusivity and efficiency is represented through accessibility, usability and security specifications that align with TAMU IT standards.

**Requirements Construct** Requirements are expressed in clear, testable terms that define what the EMS shall achieve rather than how it will be implemented. Each requirement includes purpose, rationale, constraints and verification criteria. For instance, "The system shall allow students to register for an event in 2 minutes and receive automatic confirmation" is a complete construct that includes performance and usability constraints.

**Characteristics of Individual Requirements** Every requirement must be:

- **Necessary** – Essential to meeting stakeholder or institutional objectives.
- **Feasible** – Technically and operationally achievable within the defined scope.
- **Verifiable** – Able to be confirmed by inspection, test or demonstration.
- **Unambiguous** – Interpreted consistently by all reviewers.

- **Traceable** – Linked to business, stakeholder and system specifications.

**Characteristics of a Set of Requirements** Collectively, the EMS requirements must be complete, consistent and non-conflicting. They should describe all necessary behavior of the system including functional workflows, data management, analytics and user interaction while maintaining alignment with university policies and technical standards.

**Requirement Language Criteria** The language used shall be precise, concise and active. Phrases such as "the system shall" and "the user shall be able to" will be consistently applied. Avoid vague terms like "easy" or "fast" unless accompanied by measurable metrics (e.g., "response time < 2 seconds").

**Requirements Attributes** Each requirement shall be documented with the following attributes:

- Identifier (unique ID)
- Description (clear statement of need)
- Rationale (why it exists)
- Priority (based on business value and risk)
- Verification Method (test, inspection or analysis)
- Source/Stakeholder Owner (who raised it)

## 5.2 Practical Considerations

### 5.2.1 Application of Iteration and Recursion

Requirements will be developed iteratively and refined through recurring validation cycles with CMIS stakeholders. Early prototypes, pilot events and user feedback sessions will be used to validate assumptions and adjust specifications before full deployment.

### 5.2.2 Iteration and Recursion in Requirements Engineering

The EMS requirements process will follow a recursive model where business, stakeholder, system and software requirements are interlinked. Changes in any level (e.g., a new compliance guideline or sponsor requirement) will propagate through controlled updates to maintain alignment across BRS, StRS, SyRS and SRS documents.

## 5.3 Requirement Information Items

The EMS project shall produce four key information items: Business Requirements Specification (BRS), Stakeholder Requirements Specification (StRS), System Requirements Specification (SyRS) and Software Requirements Specification (SRS). These documents will ensure traceability from institutional goals to functional implementation and serve as living assets maintained under version control throughout the project lifecycle.

# 6. PROCESSES

## 6.1 Requirement Processes

The CMIS Engagement Platform project shall implement the following three principal requirements engineering processes, as mandated by the standards cited (ISO/IEC/IEEE 15288 and 12207), which are critical for defining, structuring, and coordinating the requirements for the Event Management System and its prioritized dependencies:

| **Standard Process** | **CMIS Project Focus and Scope** |
|---|---|
| Business or Mission Analysis Process | This process will validate the strategic necessity of the platform. It will define the required change from the "patchwork of tools" to a centralized digital engagement platform to sustain CMIS's momentum and manage its complexity and growth. This includes confirming the scope of the Event Management System as a Highest Priority module. |
| Stakeholder Needs and Requirements Definition Process | This process is focused on translating the stated needs of the key stakeholders into functional needs for the Event Management System. This includes defining requirements for online registration, waitlist management, and feedback forms, addressing the student confusion over slots and lack of confirmation emails. |
| System [System/Software] Requirements Definition Process | This process will formalize the detailed, verifiable requirements for the Event Management System and its related data flows, ensuring adherence to constraints. This includes Functional Requirements – Detailed specifications for Student Signups and record maintenance, Speaker/Venue availability tracking, and Event Registration process logic. Non-Functional Requirements: Specifications for Hybrid support, Data Integrity, Usability, and Event Analytics. Security/Compliance: Ensuring requirements for data security/privacy are met, particularly concerning student data (FERPA compliance). |

### 6.1.1 Guidelines for Processes

The requirements processes for the Event Management System will be elaborated upon with the following implementation guidance to provide additional detail specific to CMIS operations:

**Event Scheduling and Confirmation:** This process is critical for addressing the existing confusion over registration and slots, as highlighted by student feedback. Requirements shall ensure the system provides a definitive, real-time status:

- **Real-Time Slot Management:** The system must automatically track capacity and update available slots for events and workshops to prevent overbooking and confusion.
- **Automated Confirmation:** Upon successful online registration, the system must immediately generate and deliver a confirmation RSVP email with event details and ticket/access

information, replacing the current failure of confirmation and addressing the student's need.
- **Waitlist Automation:** The system must implement robust waitlist management that automatically notifies and promotes users when a slot becomes available.

**Financial Tracking Integration:** Requirements for transactional activities must ensure detailed tracking for the system's financial aspects, which replaces manual compilation. This includes defining data fields and flows for:

- **Inflows and Outflows:** Tracking all money related to events, including sponsor payments, donations (alumni), and expenditures (e.g., caterers, speakers).
- **Billing Attribution:** Identifying which entity (e.g., a specific INDUSTRY_PARTNER or a specific event budget) is responsible for a bill or transaction.

**Engagement Metrics & Reporting:** The event process must capture detailed requirements for generating the necessary output reports that administrators currently lack. This includes defining the data required for Student and speaker check-in attendance (replacing manual tracking) and all necessary engagement reports (e.g., for sponsor impact).

**Quality Assurance Integration (HITL):** Requirements for data input and validation must incorporate Dr. Richards's specific mandate for a Human-In-The-Loop (HITL) process to monitor and check every 10th entry for accuracy.

## 6.2 Business or Mission Analysis Process

### 6.2.1 Purpose

The purpose of the Business or Mission Analysis Process for the EMS is to ensure that university events are seamlessly organized on a centralized platform and that the system supports CMIS's mission to centralize and digitize event operations, improving efficiency, engagement, and data integrity across all stakeholder groups.

### 6.2.2 Outcomes

Key outcomes include:

- A clear understanding of the problem: space fragmented event data and manual coordination.
- Identification of operational goals: efficiency, reliability, compliance, and stakeholder satisfaction.
- A defined system boundary linking EMS with CMIS student profiles, analytics dashboard, and sponsor portals.

### 6.2.3 Activities and Tasks

- Conduct stakeholder workshops to elicit pain points and desired outcomes.
- Map current processes (Google Forms, spreadsheets, manual confirmations) and define target workflows.
- Validate assumptions about hybrid event support and data integration with CMIS analytics.

- Establish business metrics: registration reliability, time-to-report, and budget adherence.

## 6.3 Stakeholder Needs and Requirements Definition Process

### 6.3.1 Purpose

To define and prioritize the explicit and implicit needs of all EMS stakeholders, ensure the solution is inclusive, usable, and aligned with CMIS's strategic objectives.

### 6.3.2 Outcomes

- Stakeholder needs captured through interviews, focus groups, and surveys.
- Documented and validated functional and non-functional requirements.
- Prioritized backlog of requirements with associated acceptance criteria.

### 6.3.3 Activities and Tasks

- Identify and categorize stakeholders: Students, Faculty, Industry Partners, Administrators, and Sponsors.
- Capture their event lifecycle touchpoints registration, feedback, sponsorship, and analytics.
- Translate needs into structured requirements using user stories and process models.
- Validate completeness and feasibility via stakeholder reviews and pilot prototypes.

## 6.4 System / Software Requirements Definition Process

### 6.4.1 Purpose

To transform stakeholder requirements into a detailed and implementable system specification that defines the architecture, interfaces, and constraints for the EMS solution.

### 6.4.2 Outcomes

- A complete and traceable System Requirements Specification (SRS) aligned with CMIS architecture.
- Definition of data interfaces with student profiles, analytics dashboard, and sponsor portals.
- Clear mapping between user stories, system requirements, and test cases.

### 6.4.3 Activities and Tasks

- Develop the logical architecture for the EMS using AWS reference models.
- Define APIs and integration mechanisms with CMIS subsystems.
- Specify data validation, FERPA compliance controls, and audit mechanisms.
- Develop UML diagrams, DFDs, and ERDs to visualize system interactions.
- Conduct design reviews and update requirements traceability matrix (RTM).

## 6.5 Requirements Engineering Activities in Other Technical Processes

### 6.5.1 Requirements Activities in Architecture Definition

The EMS architecture is derived directly from the system requirements. Requirements guide component modularization, ensuring scalability, data integrity, and compliance. Architectural trade-off analyses will document cost-performance decisions under the $1M–$2M grant constraint.

### 6.5.2 Requirements Activities in Verification

Verification ensures that each EMS requirement is implemented correctly. Test cases will be linked to requirements in the RTM. Key metrics include registration confirmation accuracy, system uptime, and data synchronization reliability.

### 6.5.3 Requirements Activities in Validation

Validation confirms that the EMS meets stakeholder expectations. Pilot events and simulated registration runs will confirm usability, performance, and report generation accuracy. Feedback will guide iterative refinements before full rollout.

## 6.6 Requirements Management

### 6.6.1 Management Overview

The EMS requirements management process governs how requirements are documented, versioned, traced, and controlled throughout the project lifecycle. It ensures consistency, transparency, and accountability under CMIS governance.

### 6.6.2 Change Management

Any modification to EMS requirements must pass through a formal Change Control Board (CCB) which would include the administrators' review. The process includes submission of the issues faced, impact assessment of the change, approval, and implementation tracking. This commitment ensures that necessary project changes which might involve high-priority modules like the Event Management System or addressing time-sensitive constraints like FERPA compliance are quickly evaluated and either approved or rejected, supporting the goal of Fast Deployment. Changes affecting budget, FERPA compliance, or system interoperability require escalation to Dr. Richards.

### 6.6.3 Measurement for Requirements

Measurement ensures that EMS requirements remain relevant, feasible, and verifiable. Metrics will include:

- Requirement volatility index (percentage of changed requirements per phase).
- Traceability coverage (percentage of requirements linked to test cases).
- Verification success rate (percentage of test cases passed per build).
- Stakeholder satisfaction index (from feedback surveys).

These measurements will be tracked in the Requirements Traceability Matrix (RTM) and summarized in weekly reports to Dr. Richards to ensure alignment with project goals and funding requirements.

## 6.7 Event Management System Context DFD

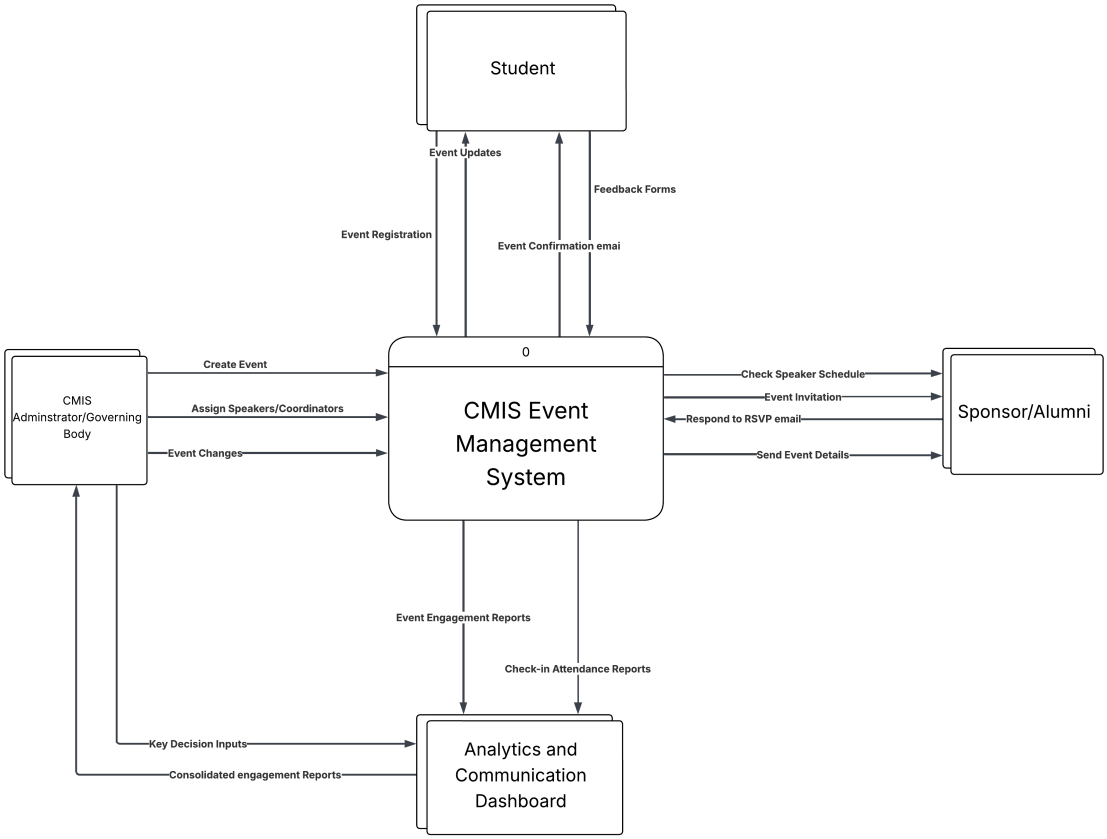



Figure 1: Event Management System Context Diagram

## 6.8 Event Management System ERD

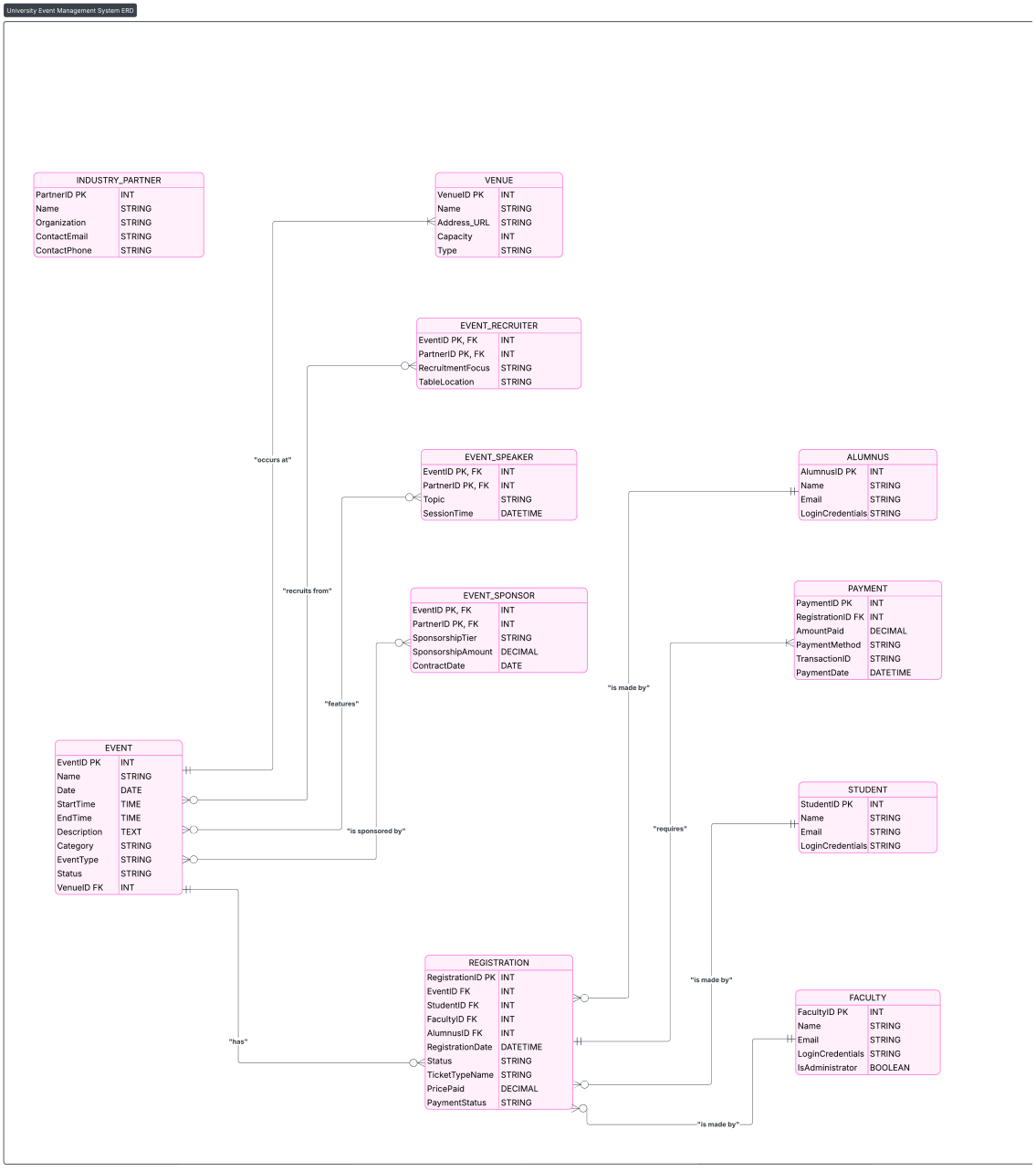



Figure 2: Event Management System Entity Relationship Diagram

# 7. INFORMATION ITEMS

The CMIS Event Management System (EMS) project shall produce the following information items in accordance with the requirements engineering framework defined by ISO/IEC/IEEE 29148:2018. These information items ensure that institutional needs, stakeholder expectations, and system behaviours are documented, traceable, and verifiable throughout the system lifecycle.

## 7.1 Information Deliverables

**Business Requirements Specification (BRS)** Captures the organizational purpose behind developing the EMS platform. It defines how the system supports CMIS's mission to simplify planning, coordination, and tracking of academic and corporate events, aligning with TAMU's digital-transformation and inclusivity initiatives. The BRS highlights key challenges in current processes – such as fragmented scheduling, manual approvals, and limited analytics and describes the expected business outcomes, including transparency, automation, and enhanced sponsor engagement.

**Stakeholder Requirements Specification (StRS)** Documents requirements from the perspective of students, sponsors, faculty coordinators, and administrative staff. It defines user expectations for event creation, room booking, sponsor integration, feedback collection, and reporting, while ensuring compliance with TAMU's accessibility, branding, and data-privacy standards.

**System Requirements Specification (SyRS)** Establishes high-level system behavior, including functional capabilities, interoperability with TAMU's enterprise ecosystem, and performance baselines. It ensures alignment with university security and single-sign-on (SSO) policies and defines how EMS will interface with existing platforms (e.g., calendar, directory, and analytics services).

**Software Requirements Specification (SRS)** Provides detailed software functionality and data structures derived from the SyRS. It describes workflows such as event creation, approval routing, registration, notifications, and feedback analysis – all within the scope of TAMU's enterprise governance and branding guidelines.

**Note 1:** Each specification will follow ISO/IEC/IEEE 29148 guidance for traceability, clarity, and completeness.

**Note 2:** Sub-specifications may be developed for core EMS modules – Event Creation, Sponsorship Portal, Feedback Dashboard, and Analytics.

**Note 3:** Specifications may exist digitally within CMIS's version-controlled repository, provided content remains accessible, logically organized, and compliant with university retention policy.

## 7.2 Information Management

All EMS information items shall be governed under TAMU's Information Management and Security Policies in alignment with ISO/IEC/IEEE 15288 and 12207. Configuration-management procedures will ensure end-to-end traceability among business needs, stakeholder expectations,

design artifacts, and validation results. Change requests will be version-controlled, with full audit history retained per university records-management requirements.

## 7.3 Representation and Documentation Approach

The EMS project employs a model-driven and document-based representation of requirements. All visual models – including process flows, use-case diagrams, and logical database schemas – will be integrated into a unified documentation framework ensuring consistency, accessibility, and role-based authorization.

Artifacts will:

- Adhere to CMIS branding and documentation templates,
- Support accessibility (WCAG 2.1 AA compliance), and
- Maintain traceability between textual and visual specifications.

These documents will serve as living assets, updated through iterative reviews with CMIS leads, the university's IT Office, and academic departments participating in pilot events.

# 8. GUIDELINES FOR INFORMATION ITEMS

## 8.1 Requirements Information Item Outlines

Each information item will clearly define its purpose, scope, assumptions, functional coverage, and traceability. All items must demonstrate alignment with TAMU's Strategic Plan 2030 pillars – Innovation, Collaboration, and Service Excellence and include measurable success criteria.

## 8.2 Business Requirements Specification (BRS)

The Business Requirements Specification (BRS) defines the institutional and strategic rationale for developing the CMIS Event Management System at Texas A&M University. It establishes the "why" behind the project, linking the EMS initiative to the university's strategic priorities of digital transformation, collaboration, inclusivity, and operational excellence.

The BRS frames the organizational motivation:

- TAMU currently manages events and sponsor collaborations using disparate tools, creating silos across departments and student groups.
- Manual processes lead to inconsistent data, duplicated communication, and limited insights into participation, funding, and engagement.
- The EMS seeks to create a unified, accessible, and auditable system for managing all event-related activities across CMIS and partner colleges.

The document will outline business goals such as:

- **Process Optimization** – Reduce administrative overhead in event planning and reporting.
- **Transparency** – Enable real-time visibility for faculty, sponsors, and administrators.
- **Data-Driven Decision-Making** – Provide analytics on participation, sponsorship, and outcomes.
- **Brand Compliance** – Ensure all CMIS events reflect TAMU's identity and accessibility standards.
- **Sustainability** – Digitize records and reduce redundant workflows.

The BRS also clarifies the expected value proposition: a centralized platform that supports innovation in student engagement, professional partnerships, and operational governance. It defines measurable success indicators (e.g., 40% reduction in manual approvals, 30% faster event scheduling, 100% digital compliance with TAMU standards).

By including this section, the RFP clearly demonstrates the business justification and ensures all subsequent requirements trace back to measurable organizational outcomes.

## 8.3 Stakeholder Requirements Specification (StRS)

### 8.3.1 General

The Stakeholder Requirements Specification (StRS) documents the needs, expectations, and constraints of every group directly or indirectly interacting with the CMIS EMS. It translates TAMU's institutional mission into user-centered perspectives, ensuring inclusivity, accessibility, and accountability.

Stakeholders include:

- **Students** – who will register, manage, and participate in events;
- **Faculty Advisors & Coordinators** – who will review, approve, and oversee events;
- **Sponsors & Partners** – who will propose, support, or evaluate events;
- **Administrators** – who will monitor performance metrics, feedback, and compliance;
- **IT & Accessibility Officers** – who will ensure the system aligns with TAMU's digital-accessibility, security, and privacy policies.

The StRS section outlines what each user expects the system to do, such as:

- Seamless event creation and modification workflows;
- Automated notification and approval mechanisms;
- Centralized dashboards for tracking attendance and outcomes;
- Role-based access to maintain data integrity;
- Built-in compliance with FERPA and WCAG 2.1 AA accessibility guidelines.

It also details qualitative expectations, such as usability, reliability, inclusivity, and visual consistency with the TAMU brand. Each requirement will be measurable (e.g., "students can create and publish an event in $\leq 5$ minutes") and verifiable during acceptance testing.

By explicitly defining stakeholder expectations, this section ensures early alignment across departments and prevents scope misinterpretation. It also assures evaluators that CMIS's EMS is not merely a technical tool – it is a service ecosystem centered on the TAMU community experience.

## 8.4 System Requirements Specification (SyRS)

### 8.4.1 General

The System Requirements Specification (SyRS) defines the complete technical and operational behavior of the CMIS Event Management System as a whole. It represents the "what the system shall do" – describing interactions between components, users, and external TAMU systems.

This section will:

- Present the conceptual system architecture (web portal, admin dashboard, sponsor module, analytics layer).
- Define system boundaries – what the EMS will and will not manage (e.g., registration but not university-wide payroll).
- Specify interfaces with existing TAMU platforms – identity management, calendar, email notifications, and institutional data repositories.
- Document system-level constraints: authentication policies, uptime targets, data-retention standards, and integration limits.
- Outline non-functional requirements, such as:
  - **Performance** – Support 1,000+ concurrent users during peak registration.
  - **Availability** – $\geq 99\%$ uptime excluding maintenance.
  - **Scalability** – Easily expand to include new departments or sponsors.
  - **Security** – Comply with TAMU Cybersecurity Control 29.01.03 M1.

Additionally, the SyRS will include conceptual models (use-case diagrams, system-context diagrams, and data-flow representations) that depict how different modules exchange data

within the CMIS ecosystem. It defines validation mechanisms – for example, verifying sponsor eligibility before event listing or syncing approved events to the TAMU calendar.

In the RFP, this section demonstrates a clear, structured system vision that integrates seamlessly with TAMU's digital infrastructure while remaining modular, secure, and scalable.

## 8.5 Software Requirements Specification (SRS)

### 8.5.1 General

The Software Requirements Specification (SRS) focuses on the implementation-ready layer of the EMS. It translates system-level expectations into precise, measurable software functions, ensuring a shared understanding between CMIS, TAMU IT, and the eventual development or vendor team.

The SRS defines:

- Functional components (e.g., Event Creation Module, Registration Engine, Feedback Analyzer, Dashboard Generator).
- Logical data models (entities like Students, Events, Sponsors, Sessions, Feedback).
- Workflows that describe the step-by-step logic of approvals, reminders, and analytics generation.
- Input/output specifications, such as how event details are captured, stored, and displayed.
- Validation and error-handling criteria, ensuring data integrity and user guidance.
- Security and privacy requirements, including encrypted data handling, access control, and audit trails.
- Performance benchmarks, such as response time, load handling, and data-processing efficiency.

It also documents traceability, mapping each software feature back to its parent system and stakeholder requirements, ensuring that no function is implemented without justification.

Beyond functionality, the SRS captures quality attributes (usability, maintainability, reliability) and defines how each will be verified through testing and user evaluation.

For the RFP, this section provides evaluators with confidence that the CMIS project:

- Has a structured approach to translating conceptual goals into executable software behavior;
- Maintains vendor neutrality by focusing on logic and standards, not tools;
- Supports long-term maintainability under TAMU's IT governance.

# 9. INFORMATION ITEM CONTENT

## 9.1 General

This clause states the normative content of the required information items.

**NOTE:** Information content in 9.2 is generally applicable to items of information maintained in document form.

## 9.2 General Content

### 9.2.1 Identification

Include the following identification matter:

a) title; and

b) revision notice.

| Item | Value |
|---|---|
| Title | Business Requirements Specification (BRS) for the CMIS Engagement Platform |
| Revision Notice | Version 1.0 (Initial Draft) |
| Date of Release | November 3, 2025 |
| Project Name | CMIS Engagement Platform (Digital Transformation Project) |

### 9.2.2 Front Matter

Include the following front matter:

a) a Table of Contents will be provided.
b) a List of Figures (to include the Business Overview Diagram).
c) a List of Tables (to include the Stakeholder List and BRS content matrix).

### 9.2.3 Definitions

Provide definitions for any words or phrases that have special meaning beyond normal dictionary definitions.

| Term | Definition |
|---|---|
| CMIS | Council for the Management of Information Systems. |
| Ad Hoc Tools | Fragmented, non-integrated software tools like Google Forms and spreadsheets currently used for management. |
| Engagement Platform | The proposed centralized digital system to integrate all CMIS interactions (students, faculty, industry, alumni). |
| Sponsor Touches | Metrics recording interactions between corporate sponsors and CMIS participants/events. |

### 9.2.4 References

Include a complete list of all documents referenced elsewhere, identifying each document by title, source, and specifying the sources from which the references can be obtained.

| Title | Type | Source/Note |
|---|---|---|
| ISO/IEC/IEEE 29148:2018 | Compliance | Requirements Engineering Standard. |
| CMIS Case Study 2025 | Compliance | Document defining the business problem and operational environment. |
| FERPA Privacy Laws | Compliance | Regulatory compliance framework for student records. |
| AWS Reference Architecture | Guidance | Proposed technology platform for implementation. |

### 9.2.5 Acronyms and Abbreviations

Spell out or define all acronyms and abbreviations used in the documents.

| Acronym | Definition |
|---|---|
| BRS | Business Requirements Specification |
| CMIS | Council for the Management of Information Systems. |
| EMS | Event Management System (a platform module). |
| FERPA | Family Educational Rights and Privacy Act. |
| MIS | Management Information Systems. |

## 9.3 Business Requirements Specification (BRS) Content

### 9.3.1 BRS Overview

This BRS defines the high-level requirements for the CMIS Engagement Platform, articulating the problem, organizational goals, and operational context necessary to justify and shape the required technical solution.

### 9.3.2 Business Purpose

The organization (CMIS at Mays Business School) is pursuing this system change because its current momentum and growth have created a level of complexity that ad hoc tools can no longer manage. The purpose is to implement a centralized digital engagement platform to sustain the mission and growth.

### 9.3.3 Business Scope

a) **Business Domain:** Council for the Management of Information Systems (CMIS) at Texas A&M Mays Business School.
b) **Range of Business Activities:** Includes shaping the MIS curriculum, fostering industry relevance, supporting career opportunities, managing events (Mixer, Career Fair, Case Competitions, Conference), and facilitating mentorship/scholarships.
c) **System Scope:** The platform will support all four key stakeholder groups: Students, Faculty, Industry Partners, and Alumni.

### 9.3.4 Business Overview

The primary entities are the CMIS organization (governed by two faculty co-directors, supported by staff and volunteers) and the Industry Partners (who provide financial and engagement support). The relationship is a direct bridge between academia and industry.

### 9.3.5 Major Stakeholders

The major stakeholders influencing the system development are:

- **Students:** Need clarity on workshops and integrated access to career services.
- **Faculty (Co-Directors/Professors):** Need tools for managing schedules, submissions, and accessing analytics efficiently.
- **Industry Partners (Recruiters/Sponsors):** Need consistent, streamlined recruiting processes and access to student resumes/profiles.
- **Administrators (CMIS Coordinator):** Need an easy way to produce engagement and attendance reports.
- **Alumni:** Need a central place to find mentorship opportunities and connect easily.

### 9.3.6 Business Environment

**External Factors:** Includes market trends influencing MIS curriculum relevance, and legal and regulation requirements like FERPA privacy laws.

**Internal Factors:** The tension between the increasing scope of activities and limited coordinating resources (staff and student volunteers) drives the need for automation.

### 9.3.7 Mission, Goals and Objectives

- **Mission (Core):** To create a bridge between industry and academia.
- **Goal (Sustained Growth):** To maintain the program's momentum and reputation by professionalizing operations.
- **Objective (Measurement):** Produce timely and accurate reports demonstrating sponsor ROI (impact) and student engagement data.

### 9.3.8 Business Model

The core mission is achieved through:

- **Product/Services:** Events (conferences, career fairs), curriculum shaping, and mentorship/scholarships.
- **Revenue Model:** Financial and active engagement support provided by corporate partners across various industries.

### 9.3.9 Information Environment

a) **Project Portfolio:** This platform is the primary strategic project, aiming to consolidate all operational data streams.
b) **Long Term System Plan:** The strategy is to leverage existing staff familiarity with the AWS cloud environment and utilize a low-maintenance, scalable architecture.

c) **Database Configuration:** The strategic decision is to establish a centralized digital engagement platform (Data Layer) rather than relying on distributed, non-integrated tools (spreadsheets, SurveyMonkey).

#### 9.3.10 Business Processes

The primary system interfaces must support the event lifecycle:

- **Registration Process:** Must unify the steps currently separated across Google Forms, Spreadsheets, and Email Confirmations.
- **Reporting Process:** Must automate the manual compilation of attendance, sponsor touches, and engagement data across platforms.

#### 9.3.11 Business Operational Policies and Rules

- **Tiered Access Rule:** Access to certain features, events, and student data must align with the sponsor's Sponsor Tier status (Exabyte, Terabyte, etc.).
- **FERPA/Privacy Rule:** Student information access by unauthorized staff or external parties is strictly prohibited and must require consent for disclosure.

#### 9.3.12 Business Operational Constraints

- **Technical Constraint:** The platform must be built to leverage existing staff familiarity with the AWS cloud.
- **Maintenance Constraint:** The solution must require minimal ongoing maintenance effort, as only a single MIS student worker is available for support.

#### 9.3.13 Business Operational Modes

- **Normal Mode:** Full support for online registration, real-time analytics, and automated communication.
- **Unsteady/Manual Mode (Mitigation):** Although the platform aims to eliminate manual processes, the system must retain or provide a mechanism for manual overrides for issues like last-minute speaker changes or manual attendance logging if digital tools fail.

#### 9.3.14 Business Operational Quality

The system must prioritize usability for both students (clear sign-up) and recruiters (ease of access/search). The platform must deliver data integrity to prevent lost registrations or inconsistent data, ensuring the overall reliability of operations.

#### 9.3.15 Business Structure

The system must support the existing structure of:

- **Leadership:** Two Faculty Co-Directors (Dr. Gomillion and Dr. Whitten).
- **Staffing:** Staff coordinators and a reliance on student volunteers and a dedicated student worker for maintenance.
- **Partners:** An advisory board of Industry Partners organized by tiered sponsorship (Exabyte, Terabyte).

### 9.3.16 High-level Operational Concept

The proposed system is a centralized digital engagement platform that acts as the single source of truth, eliminating fragmented data silos.

a) **Description:** The system unifies profiles (Student Career Portal), events (Event Management), and decision support (Analytics Dashboard) under a single, integrated AWS architecture.
b) **Modes of Operation:** Supports seamless, integrated operation across all major CMIS functions (recruiting, events, academics).

### 9.3.17 High-level Operational Scenarios

- **Scenario 1 (Student Registration):** A student logs into the portal and successfully registers for a workshop in real-time, receiving an immediate email confirmation, eliminating the problem of missing confirmations and confusion over capacity.
- **Scenario 2 (Recruiter Sourcing):** An Exabyte Sponsor logs into the portal, searches for seniors with specific skill tags, and instantly accesses their most recent, permission-granted resume, without requesting it manually.

### 9.3.18 Other High-level Life-Cycle Concepts

- **Acquisition/Development Concept:** The system will be built by a student team and maintained by an MIS student worker, relying on AWS infrastructure.
- **Deployment Concept:** A goal for Fast Deployment is prioritized to quickly address the operational issues mounting from the successful growth.

### 9.3.19 Project Constraints

The project operates under constraints that prioritize a cost-effective, maintainable solution:

- The system must minimize budget for operational costs.
- A goal for Fast Deployment is mandatory to quickly mitigate current operational risks.
- The final solution must accommodate minimal human resources for long-term maintenance (MIS student worker).

## 9.4 Stakeholder Requirements Specification (StRS) Content

### 9.4.1 StRS Overview

The Stakeholder Requirements Specification (StRS) defines the functional, experiential, and operational requirements of all user groups who will interact with the CMIS Event Management System (EMS). It documents stakeholder expectations, constraints, and success criteria to ensure that the proposed platform aligns with the real needs of the Council for the Management of Information Systems (CMIS) at Texas A&M University Mays Business School. The StRS converts institutional goals described in the Business Requirements Specification into measurable user-level behaviors and experience outcomes.

### 9.4.2 Stakeholder Purpose

The purpose of this specification is to make sure every key participant, including students, faculty, sponsors, administrators, and alumni, receives a unified digital experience that minimizes manual work, increases visibility into event operations, and enables data-driven decision-making. The StRS describes how the EMS will deliver value to each stakeholder group and how these expectations will influence design, usability, and data management.

### 9.4.3 Stakeholder Scope

a) **System Domain:** CMIS Event Management System (EMS) within the CMIS Engagement Platform.
b) **Scope of Influence:** All users involved in planning, executing, or evaluating CMIS events, including faculty co-directors, student organizations, sponsors, and the Mays IT team.
c) **Operational Scope:** Covers the complete event lifecycle including creation, approval, registration, execution, feedback collection, and analytics.
d) **Non-Scope Areas:** External financial systems and university-wide reservations not governed by CMIS.

### 9.4.4 Overview

The CMIS EMS is a centralized digital solution that helps faculty and students manage events efficiently while allowing sponsors and alumni to participate through structured digital access. The StRS defines expectations for user journeys, access levels, and data interactions that comply with Texas A&M University's accessibility standards (WCAG 2.1 AA) and data-privacy policies (FERPA).

### 9.4.5 Stakeholders

| Stakeholder Group | Primary Interests | Typical Interaction |
|---|---|---|
| Students | Register for events, view schedules, access feedback | Event portal and registration forms |
| Faculty (Co-Directors / Advisors) | Approve events, monitor analytics, view reports | Faculty dashboard and approvals workflow |
| Sponsors / Recruiters | Review engagement data, measure ROI, access student profiles | Sponsor portal with engagement analytics |
| Administrators | Manage records, generate reports, maintain permissions | Admin console and management tools |
| Alumni | Participate in events and mentorship activities | Integrated portal connected to the Mentorship Marketplace |
| TAMU IT and Accessibility Officers | Validate security and compliance | Access audits and compliance dashboards |

### 9.4.6 Business Environment

The EMS operates within the academic and administrative environment of Texas A&M University. It follows university branding and accessibility standards and must comply with FERPA and institutional data-governance requirements. Internally, CMIS depends on limited staff and student support, which makes automation essential for sustainable operations.

### 9.4.7 Mission, Goals, and Objectives

**Mission:** Enable seamless collaboration between academia, students, and industry partners through digital engagement.

**Goals:** Digitize all event operations, reduce manual effort by at least 70 percent, and improve reporting accuracy.

**Objectives:** Provide a role-based portal with real-time analytics, transparent communication, and automated event coordination.

### 9.4.8 Business Model

The EMS supports a service-based engagement model for CMIS. It connects academic programming, industry partnerships, and student opportunities through events and analytics. Corporate partners provide both financial and interactive support, while students and faculty supply the academic and operational framework. The system ensures measurable sponsor return on investment and long-term engagement continuity.

### 9.4.9 Information Environment

The EMS will operate as the central repository of event and engagement data for CMIS. It will consolidate existing spreadsheets, forms, and email workflows into one cloud-based database hosted on the Texas A&M AWS infrastructure. All information will be governed through role-based permissions and backed by automatic version control.

### 9.4.10 System Processes

Key stakeholder-driven processes include:

- Event Creation and Approval Workflow.
- Participant Registration and Confirmation.
- Sponsor Engagement and Feedback.
- Post-Event Reporting and Analytics.

Each process is traceable, automated, and auditable to support transparency and accountability.

### 9.4.11 System Operational Policies and Rules

- Access to specific features or data must correspond to user roles and sponsor tiers (Exabyte, Terabyte, Gigabyte).
- All student data will be protected according to FERPA guidelines and shared only with explicit consent.
- Automated workflows must retain manual override options for faculty in case of urgent or exceptional situations.

### 9.4.12 Operational Constraints

The system must operate within TAMU's cybersecurity and budgetary frameworks. It should require minimal human maintenance since only one MIS student worker is available for technical upkeep. Deployment must remain cost-effective and hosted entirely in the TAMU AWS environment.

### 9.4.13 System Operational Modes and States

- **Normal Mode:** All automated functions active including registration, scheduling, and analytics.
- **Maintenance Mode:** System accessible to administrators for updates or configuration.
- **Manual Mode:** Allows temporary offline or partial operation if automation services are unavailable.

### 9.4.14 System Operational Quality

The EMS must be reliable, accurate, and easy to use. It must maintain data integrity during high-volume registration periods and ensure quick response times. User satisfaction and accessibility compliance will be core quality indicators.

### 9.4.15 User Requirements

- Students must be able to register for events within five minutes and receive immediate confirmation.
- Faculty must approve or reject events through a single dashboard view.
- Sponsors must be able to view engagement metrics and download summary reports securely.
- Administrators must generate event attendance and ROI reports on demand.
- All users must access the system on desktop and mobile devices with full accessibility support.

### 9.4.16 Operational Concept

The EMS functions as the digital backbone of CMIS operations. It unifies event scheduling, participant registration, sponsor tracking, and analytics into a single workflow. The platform supports asynchronous participation and can adapt to hybrid or fully online event models.

### 9.4.17 Operational Scenarios

- **Scenario 1: Student Registration** – A student logs in with TAMU credentials, selects an event, registers instantly, and receives a confirmation email and calendar invite.
- **Scenario 2: Sponsor Engagement** – A corporate sponsor logs in to review event participation data, downloads a custom ROI report, and provides feedback through a sponsor portal form.
- **Scenario 3: Faculty Analytics Review** – A faculty director accesses the dashboard to analyze attendance trends and generate a quarterly impact summary for the CMIS board.

### 9.4.18 Other Detailed Concepts of Proposed System

The system must support future integration with modules such as the Mentorship Marketplace and Research Collaboration Hub. It will include audit logging and data export features to enable institutional research and performance assessment.

### 9.4.19 Project Constraints

- The system must be implemented using resources available through CMIS and TAMU IT without dedicated external vendors initially.
- Development and testing must be completed within the academic calendar cycle to support Fall 2026 events.
- Maintenance tasks must remain manageable by a student worker supervised by faculty advisors.
- All development work must comply with university data-governance and cloud-hosting policies.

## 9.5 System Requirements Specification (SyRS) Content (Event Management System)

### 9.5.1 SyRS Overview

This document specifies the requirements for the CMIS Event Management System (EMS), a single, unified, cloud-hosted platform. The project shall produce this content to transition CMIS from a manual coordination model to a proactive, intelligence-driven digital ecosystem, aligning with the project's policies for system effectiveness and measurable value creation.

### 9.5.2 System Purpose

The reason for developing the EMS is to eliminate structural inefficiencies caused by the absence of an integrated event management solution. The system's core purpose is to consolidate every stage of event planning, execution, and evaluation, and to transition CMIS from a reactive, manual coordination model to a proactive, intelligence-driven digital ecosystem.

### 9.5.3 System Scope

a) **System Name:** CMIS Event Management System (EMS).
b) **Needs Analysis/Problem Expression:** The current model uses a fragmented combination of email threads, spreadsheets, and form-based tools for administrative tasks, resulting in structural inefficiencies. This disjointed approach hinders the Council's ability to manage large-scale events effectively and prevents leveraging data for institutional decision-making. The EMS will satisfy these needs by establishing a single, unified environment that automates repetitive workflows.
c) **Application Description/Benefits:** The EMS will serve as the digital backbone of CMIS operations. Top-level benefits include:
    - Reducing administrative overhead by at least 70% via automation.
    - Enabling leadership to monitor event ROI, engagement patterns, and participation trends.
    - Fostering continuous collaboration, knowledge exchange, and alumni engagement.

### 9.5.4 System Overview

**System Context** The EMS is a single cloud-hosted platform that unifies registration, scheduling, and analytics. It serves as the digital backbone of CMIS operations.

Significant Interfaces Crossing the Boundaries:

- **Data Input:** Student profiles for registration.
- **Data Output:** Interactive dashboards powered by AWS QuickSight.
- **Future Integration:** Mentorship Marketplace and Research Collaboration Hub.

**System Functions** Major system capabilities include:

- **Integration and Automation:** Automating repetitive workflows like confirmation emails, attendance tracking, and report generation.
- **Data-Driven Insight:** Transforming raw participation data into actionable intelligence.
- **Stakeholder Empowerment:** Providing tailored experiences via role-based dashboards.

**User Characteristics** The system must provide tailored experiences for the following user types:

| User Type | Nature of Use | Characteristics/Capabilities |
|---|---|---|
| Students | Event registration, session scheduling, accessing personalized experiences. | Lack a single point of interaction; face repetitive registrations. |
| Faculty/Staff/Admins | Event planning, session scheduling, tracking attendance, logistics, consolidating feedback. | Currently burdened by manual workflows. |
| Sponsors/Alumni | Communication, event engagement, viewing engagement data, accessing tailored experiences. | Sponsors have limited visibility into engagement data or student metrics. |
| Leadership | Monitoring event ROI, long-term participation trends, institutional decision-making. | Need easy monitoring and evidence-based reporting. |

### 9.5.5 Functional Requirements

The system shall define functionalities that consolidate administrative tasks:

- The system shall provide online registration that eliminates repetitive data entry for students.
- The system shall automate workflows for confirmation emails, attendance tracking, and report generation.
- The system shall enable the scheduling of sessions and coordination of logistics.
- The system shall provide a mechanism for sponsor communication and feedback collection.
- The system shall implement role-based dashboards for students, faculty, sponsors, and alumni.

### 9.5.6 Usability Requirements

- The system shall ensure usability across devices.
- The system shall provide tailored experiences for all stakeholder groups.
- The system shall maintain consistency and clarity to achieve high stakeholder satisfaction.

### 9.5.7 Performance Requirements

- **Response Time:** The system shall provide real-time access to participation data.
- **Efficiency Criteria:** The system must meet the objective to reduce administrative overhead by at least 70%.
- **Endurance:** The system must be scalable to manage large-scale events effectively and support the continued expansion of corporate partnerships and student engagement initiatives.

### 9.5.8 System Interface Requirements

- The system shall establish a single interface that unifies registration, scheduling, and analytics.

- **External Entity Interface (Analytics):** The system shall interface with AWS QuickSight to power interactive dashboards.
- **External Entity Interface (Compliance):** The system must adhere to WCAG 2.1 standards for accessibility and usability.
- **System Element Interface (Future):** The EMS must be designed for integration with future modules such as the Mentorship Marketplace and Research Collaboration Hub.

### 9.5.9 System Operations

**Human System Integration Requirements**

- The system must minimize the need for extensive manual reconciliation in tasks like attendance tracking and consolidating post-event feedback.
- The system shall automate repetitive workflows to reduce the significant faculty and student time currently consumed by manual processes.

**Maintainability Requirements**

- The system must transition to a proactive, intelligence-driven digital ecosystem to avoid the current reactive, manual coordination model.
- The system shall ensure that data fragmentation does not occur, avoiding the need for extensive maintenance to merge data across multiple systems.

**Reliability Requirements**

- The system reliability must ensure that it can manage large-scale events effectively.
- The system must demonstrate data integrity to minimize the risk posed by the current inefficiencies.

**Other Quality Requirements**

- **Compatibility:** The system must ensure accessibility and usability across devices.
- **Quality Standard:** The system must align with ISO/IEC/IEEE 29148 standards for system effectiveness and measurable value creation.

### 9.5.10 System Modes and States

- **Operational Mode:** Full functionality available, actively consolidating every stage of event planning, execution, and evaluation.
- **Analytical Mode:** Data is actively being transformed into actionable intelligence and used to support predictive analytics and longitudinal reporting.
- **Sustainment Mode:** The platform acts as a living repository of event intelligence.

### 9.5.11 Physical Characteristics

**Physical Requirements**

- The system shall not impose constraints on weight, volume, or dimension, as it is designed as a cloud-hosted platform.
- The system must ensure accessibility and usability across devices.

**Adaptability Requirements**

- The system must be built with adaptability to support future integration with future modules such as the Mentorship Marketplace and Research Collaboration Hub.
- The system must demonstrate scalability to support the continuous expansion of corporate partnerships and student engagement initiatives.

### 9.5.12 Environmental Conditions

The EMS is not subject to physical environmental conditions.

- **Business Environment:** The system must support the CMIS's goal of demonstrating measurable impact to external sponsors and internal stakeholders.
- **Legal/Regulatory Environment:** The system must adhere to WCAG 2.1 and FERPA compliance requirements.

### 9.5.13 System Security Requirements

- The system must protect against unauthorized access or disclosure of information related to sponsor engagement levels or student participation metrics.
- Security must ensure adherence to FERPA compliance requirements.
- Access limitations must be enforced via role-based dashboards.

### 9.5.14 Information Management Requirements

- The system shall unify registration, scheduling, and analytics, preventing data fragmentation across multiple systems.
- The system shall ensure synchronization and version control.
- The system shall act as a living repository of event intelligence, supporting longitudinal reporting and predictive analytics.

### 9.5.15 Policy and Regulation Requirements

- **Privacy Regulation:** The system must adhere to FERPA compliance requirements.
- **Accessibility Regulation:** The system must adhere to WCAG 2.1 standards.
- **Organizational Policy:** The system must align with Texas A&M University's broader digital transformation agenda.

### 9.5.16 System Life Cycle Sustainment Requirements

- The system must be sustained to allow the transformation of participation data into actionable intelligence over time.
- The platform must be maintained as a living repository of event intelligence to support future predictive analytics.

### 9.5.17 Packaging, Handling, Shipping and Transportation Requirements

These requirements are not applicable as the EMS is a single, unified, cloud-hosted platform.

### 9.5.18 Verification

Verification approaches shall be planned to qualify the system against the following:

- **Functional (9.5.5):** Verification of automation functionality to confirm the reduction in administrative overhead by at least 70%.
- **Usability (9.5.6):** Verification of usability across devices and adherence to WCAG 2.1.
- **Performance (9.5.7):** Verification that the system can manage large-scale events effectively and provide real-time data.
- **Security (9.5.13):** Verification of FERPA compliance and the security of role-based dashboards.

## 9.6 Software Requirements Specification (SRS) Content (Focus: Mentorship Marketplace)

### 9.6.1 SRS Overview

This document specifies the software requirements for the Mentorship Marketplace (MM), a Secondary Priority module. It is intended to provide a structured platform for connecting students with alumni and industry partners, supporting the platform's long-term goal of fostering continuous collaboration, knowledge exchange, and alumni engagement.

### 9.6.2 Purpose

The purpose of the MM software is to formalize and structure mentorship opportunities, transitioning this interaction from an informal (word-of-mouth) process to an integrated, manageable system, thereby promoting alumni engagement.

### 9.6.3 Scope

a) **Software Product Name:** Mentorship Marketplace (MM) Software Module.
b) **What the Software Will Do:** The MM will provide tools for profile creation, preference matching (mentor/mentee), internal communication, scheduling, and tracking engagement success.
c) **Benefits, Objectives, and Goals:** The MM aims to achieve CMIS's long-term goal of fostering continuous collaboration, knowledge exchange, and alumni engagement.
d) **Consistency:** This is a module intended for integration with future modules.

### 9.6.4 Product Perspective

The MM is an element of the larger CMIS Engagement Platform.

**System Interfaces** The MM will interface with:

- **Identity & Access (Secure Login):** For authenticating students and alumni/sponsors as mentors.
- **Student Career Portal (Data Input):** To access student profile data (skills, interests) for the matching algorithm.
- **Analytics & Comms Dashboard (Data Output):** To provide data on successful matches, meeting frequency, and mentor engagement for longitudinal reporting.

**User Interfaces** The system shall utilize a unified web-based interface consistent with the EMS and other modules, providing tailored experiences through specific mentor and mentee dashboards.

**Software Interfaces** The MM software will interface with the platform's Messaging layer to handle automated reminders (e.g., meeting cadence suggestions) and internal communication.

**Communications Interfaces** The MM requires secure, internal communication channels (e.g., a messaging tool) to facilitate contact between mentors and mentees, ensuring data privacy and supporting structured interaction.

**Memory Constraints** The MM must be scalable to store and process a growing repository of mentor and mentee profiles and conversation histories, supported by the single cloud-hosted platform architecture.

### 9.6.5 Product Functions

The major functions are:

- **Profile Creation:** Allow students and alumni/sponsors to create profiles, listing skills, interests, and mentorship preferences.
- **Matching Algorithm:** Functionality to suggest pairings based on defined criteria (e.g., industry, major, graduation year).
- **Communication and Scheduling:** Tools for internal messaging and suggesting meeting cadence for matched pairs.
- **Capacity Management:** Ability for mentors to set limits on the number of mentees they can accept to prevent being overloaded.

### 9.6.6 User Characteristics

| User Group | Characteristics/Influence | Reason for Requirements |
|---|---|---|
| Students (Mentees) | Seeking specific career guidance and connections. | Need a single, accessible platform (not word-of-mouth). |
| Alumni/Sponsors (Mentors) | Time-constrained professionals, often high-level executives. | Need capacity controls to prevent being overloaded. |
| Administrators | Overseeing program success and growth. | Need data for predictive analytics and longitudinal reporting. |

### 9.6.7 Limitations

a) **Regulatory Requirements:** Must adhere to FERPA compliance requirements for student profile data used in matching.
b) **Safety and Security Considerations:** The design must protect the privacy of interactions and the confidentiality of personal data shared within the mentorship context.

### 9.6.8 Assumptions and Dependencies

- **Dependency:** The module is dependent on the Alumni portal (or similar system) providing accurate, opted-in profiles for potential mentors.
- **Assumption:** Mentors and mentees are willing to engage with meeting cadence suggestions provided by the system.

### 9.6.9 Apportioning of Requirements

The MM module is a Secondary Priority feature intended to be implemented after the three Highest Priority modules (EMS, Student Career Portal, Case Competition Platform).

### 9.6.10 Specified Requirements

- The system shall implement an algorithm to match student and mentor profiles based on defined preferences and availability.

- The system shall provide an internal communication tool that respects the privacy of interactions.

### 9.6.11 External Interfaces (Detailed Data Flows)

| Interface Item | Purpose | Source/Destination | Data Format | Timing |
|---|---|---|---|---|
| Profile Data for Matching | Provides initial data for the matching algorithm. | Source (Student Career Portal / Alumni Portal) / Destination (MM DB) | JSON/API Endpoint | On user profile update/match initiation |
| Match Status Record | Tracks the outcome and activity of a mentor-mentee pairing. | Output (MM DB) / Destination (Analytics & Comms Dashboard) | JSON/Parquet | Real-time on status change or scheduled feed |
| Meeting Reminder | Automated communication to enforce suggested meeting cadence. | Output (MM Messaging Layer) / Destination (User Email) | SMTP/Internal Notification | As per scheduled cadence suggestion |
| Mentor Capacity Update | Provides real-time data on mentor availability. | Input (Mentor Dashboard) / Destination (MM DB) | Boolean/Integer API | Real-time on mentor update |

### 9.6.12 Functions

a) **Validity Checks:** The capacity function must prevent a mentor from being matched if their set limit is reached.
b) **Exact Sequence of Operations:** The system runs the Matching Algorithm, checks capacity, sends an invitation, and upon acceptance, initiates meeting cadence suggestions.
c) **Responses to Abnormal Situations:** If the internal messaging function fails, the system must log the error and notify the user to ensure continuous communication.

### 9.6.13 Usability Requirements

- The system shall achieve high user satisfaction by making the profile creation and preference setting simple and quick for both mentors and mentees.
- The interface must clearly indicate when a mentor's capacity is full to manage expectations and avoid overloading mentors.

### 9.6.14 Performance Requirements

- The matching algorithm must process and return suggested pairings in a timely manner (e.g., less than 10 seconds).
- The internal messaging functionality must support real-time communication.

### 9.6.15 Logical Database Requirements

The database must logically store the following entities and relationships:

- Mentor Profile and Mentee Profile (linked to the Student Career Portal).

- Match Records, including match criteria and status (active/inactive).
- Message/Communication History (for security and audit purposes).
- Integrity constraints to enforce mentor capacity limits.

### 9.6.16 Design Constraints

- The system design must incorporate mechanisms to manage and display mentor capacity to avoid overloading mentors.
- The software architecture must be simple to maintain, aligning with the goal of low maintenance by a student worker.

### 9.6.17 Standards Compliance

The software must adhere to:

- FERPA compliance requirements (for data privacy).
- WCAG 2.1 standards for accessibility.

### 9.6.18 Software System Attributes

a) **Reliability:** The matching algorithm must be reliable and consistently produce relevant, high-quality pairings.
b) **Security:** Requirements to protect the privacy of interactions and the confidentiality of mentor/mentee communication.
c) **Maintainability:** The software must be modular to facilitate future updates, supporting its role as part of the living repository of event intelligence.

### 9.6.19 Verification

Verification approaches shall be planned to qualify the MM software against its core functional and security requirements.

| **Requirement ID/Area** | **Verification Method** | **Pass/Fail Criteria** |
|---|---|---|
| Reliability/Matching | Test (T) (Unit/Integration Testing) | The Matching Algorithm must consistently produce $\geq 3$ relevant suggestions based on a set of 10 predefined test profiles. |
| Usability (Capacity Management) | Demonstration (D) (UAT with Admin/Alumni) | A mentor must be able to change their capacity limit in $\leq 3$ clicks, and the system must immediately reflect this change on the mentee's view. |
| Security/Privacy | Analysis (A) (Security Audit) | Successful adherence to FERPA compliance requirements and verification that the internal messaging tool isolates conversations to only matched pairs, protecting the privacy of interactions. |

# ANNEX A – SYSTEM OPERATIONAL CONCEPT (NORMATIVE)

This annex defines the operational concept of the CMIS Event Management System and explains how it functions within its intended environment at Mays Business School. It describes the workflows, interactions, and processes that support event creation, approval, registration, participation, and reporting. The concept outlines how administrators, faculty coordinators, students, and sponsors collaborate through a unified digital platform to ensure smooth coordination and accurate communication. It highlights how each user group engages with the system to complete tasks efficiently while maintaining compliance with university standards and data governance policies. The annex also emphasizes the system's role as the central digital backbone connecting CMIS initiatives and supporting seamless data flow between key modules such as the sponsorship portal, registration dashboard, and analytics reports.

# ANNEX B – CONCEPT OF OPERATIONS (INFORMATIVE)

This annex defines the overall concept of operations for the CMIS Event Management System and describes how it will be used to fulfill the institutional mission of CMIS. It explains how the system supports the entire event lifecycle from initial planning and scheduling to execution, feedback collection, and post event analysis. The concept includes real world examples such as student self registration, faculty approval workflows, automated notifications, and real time analytics for decision making. It also describes system behavior during both normal and exceptional conditions including large scale career fairs and hybrid academic events. Dependencies on Texas A&M University infrastructure such as authentication, cloud storage, and accessibility compliance are also addressed. This annex provides a clear operational picture for stakeholders to understand how the system aligns with university processes and contributes to CMIS goals.

# ANNEX C – TAILORING POLICIES (NORMATIVE)

This annex defines the project's tailoring approach to international standards including ISO, IEC, and IEEE 29148 and 15288. It outlines how CMIS will adapt these standards to suit the scale and nature of the project. Tailoring policies specify which requirements, documentation formats, and verification activities are mandatory and which can be adjusted for academic context. The focus is on maintaining traceability, version control, and quality assurance while allowing flexibility in documentation depth and tool selection. It ensures consistency across all requirement specifications including Business, Stakeholder, System, and Software and establishes procedures for managing updates, audits, and approvals within CMIS governance.

# BIBLIOGRAPHY

The bibliography lists all documents and references that informed the development of the Event Management System. It includes ISO, IEC, and IEEE standards for systems and software engineering, Texas A&M University data governance and accessibility policies, FERPA regulations, and AWS architecture guidelines used in defining the system's cloud infrastructure. It also cites CMIS internal documents such as case studies, user feedback reports, and previous event performance summaries. These references provide a foundation for maintaining technical accuracy, compliance, and academic rigor throughout the project documentation.

## Standards and Regulations


- ISO/IEC/IEEE 29148:2018, Systems and software engineering – Life cycle processes – Requirements engineering
- ISO/IEC/IEEE 15288:2015, Systems and software engineering – System life cycle processes
- ISO/IEC/IEEE 12207:2017, Systems and software engineering – Software life cycle processes
- Family Educational Rights and Privacy Act (FERPA), 20 U.S.C. §1232g
- Web Content Accessibility Guidelines (WCAG) 2.1 Level AA


## University Policies and Guidelines


- Texas A&M University Information Technology Governance Framework
- Texas A&M University Data Security and Privacy Policy
- Texas A&M University Cybersecurity Control 29.01.03 M1
- Texas A&M University Strategic Plan 2030
- Mays Business School Branding and Accessibility Standards


## Technical References


- Amazon Web Services (AWS) Reference Architecture for Educational Institutions
- AWS QuickSight Documentation and Best Practices
- Cloud Infrastructure Design Patterns for Higher Education


## Internal Documents


- CMIS Case Study 2025 – Operational Environment and Business Problem Definition
- CMIS Event Performance Summary Reports (2023-2025)
- CMIS Stakeholder Feedback Collection and Analysis
- Mays Business School IT Infrastructure Assessment

# IEEE NOTICES AND ABSTRACT

## IEEE Notices

This document has been prepared in accordance with IEEE standards and guidelines for systems and software requirements engineering. All references to IEEE, ISO, and IEC standards are used for the purpose of ensuring technical accuracy, compliance, and academic rigor. The intellectual property and usage rights associated with these standards remain with their respective organizations.

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## Abstract

The CMIS Event Management System Requirements Documentation defines the comprehensive requirements for developing a secure, data-driven, and user-friendly platform that enhances collaboration among students, faculty, sponsors, and alumni at Mays Business School, Texas A&M University. The system addresses current operational inefficiencies by providing a unified digital environment that manages the complete event lifecycle from planning and registration to execution and analytics.

This documentation encompasses Business Requirements Specification (BRS), Stakeholder Requirements Specification (StRS), System Requirements Specification (SyRS), and Software Requirements Specification (SRS), ensuring full traceability from institutional goals to technical implementation. The platform is designed to automate repetitive workflows, enable real-time decision-making through advanced analytics, and maintain compliance with university standards including FERPA privacy requirements and WCAG 2.1 accessibility guidelines.

By investing in this integrated event management solution, CMIS strengthens its mission to build connections between academia and industry, streamline operations, and deliver a best-in-class digital experience that embodies the values of excellence, leadership, and innovation characteristic of Texas A&M University. The system serves as the digital backbone for CMIS operations and supports the university's broader vision of digital transformation and continuous improvement in student engagement and professional development.